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CampusInsight AI: Institutional Performance Platform

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ABSTRACT

- CampusInsight AI uses Agentic AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Explainable AI (XAI) to analyze institutional accreditation documents and identify quality gaps.
- The proposed system generates explainable recommendations, interactive dashboards, and automated reports to improve institutional quality and accreditation readiness.

INTRODUCTION

- Higher Education Institutions generate SSR, AQAR, departmental reports, and other quality assurance documents for NAAC accreditation.
- Manual evaluation of these documents is time-consuming and may overlook quality gaps and missing evidence.
- Agentic AI, Large Language Models (LLMs), and Explainable AI (XAI) enable intelligent and transparent document analysis.
- CampusInsight AI uses multiple AI agents to analyze documents, identify quality gaps, and generate explainable recommendations.
- The proposed system aims to improve document quality, reduce manual effort, and enhance accreditation readiness.

PROBLEM DEFINITION

- To design and develop an Agentic AI-based institutional document intelligence platform that automatically analyzes institutional documents such as SSR, AQAR, and departmental reports, identifies quality gaps and missing evidence, generates explainable recommendations, and supports institutions in improving document quality and accreditation readiness.

SCOPE OF THE SYSTEM

- Automates the analysis of SSR, AQAR, and other institutional quality documents.
- Detects quality gaps, missing evidence, and NAAC compliance issues using Agentic AI and Large Language Models (LLMs).
- Generates Explainable AI (XAI)-based recommendations to improve accreditation readiness.
- Provides dashboards, analytics, and automated reports for Administrators, HODs, Faculty Members, and Accreditation Committees.
- Supports continuous institutional quality improvement and efficient NAAC accreditation preparation.

OBJECTIVE

- To develop an Agentic AI-based platform for intelligent analysis of institutional quality documents.
- To automate document processing, quality assessment, and NAAC criteria mapping using Artificial Intelligence.
- To detect quality gaps and missing evidence through multi-agent AI analysis.
- To generate Explainable AI (XAI)-based recommendations and comprehensive accreditation reports.
- To improve NAAC accreditation readiness while reducing manual effort and decision-making time.

• [1] A Multi-Agent RAG Framework for Regulatory Compliance

Checking of Software Requirements

- **Authors:** Souvick Das, Novarun Deb, Nabendu Chaki and Agostino Cortesi
- **Published in:** ACM Transactions on Software Engineering and Methodology (TOSEM), July 2026
- **Methodology:**
 - Implemented using **Python**.
 - Utilizes **Multi-Agent Retrieval-Augmented Generation (RAG)**.
 - Integrates **Large Language Models (LLMs)** for document analysis.
 - Employs **Explainable AI (XAI)** to generate transparent recommendations.
 - Detects compliance issues through intelligent document analysis.
- **Accuracy:** Evaluated using Context Precision, Context Recall, Answer Relevancy, and Faithfulness metrics, demonstrating effective compliance checking performance.
- **Advantages:**
 - Automated compliance analysis using Multi-Agent RAG.
 - Transparent AI decisions through Explainable AI (XAI).
- **Disadvantages:**
 - High computational complexity.
 - Requires multiple AI models and external knowledge retrieval.

- [2] **Alvaluate: A Multi-Agent Framework for Automated Answer Script Evaluation Using Large Language Models and Semantic Vector Indexing**
 - **Authors:** P. Suryakumar, A. Malini, S. K. Subasini, G. Priyanka and V. Sandhiya
 - **Published in:** IEEE Access, September 2025
 - **Methodology:**
 - Implemented using **Python**.
 - Utilizes **OCR, Multi-Agent AI, FAISS**, and **GPT-4o-mini**.
 - Automatically evaluates descriptive answer scripts and generates feedback.
 - **Accuracy:** Experimental results show strong correlation between AI-generated scores and expert evaluation while reducing evaluation time by approximately **70–80%**.
 - **Advantages:**
 - Faster and consistent answer script evaluation.
 - Reduces manual effort through AI automation.
 - Generates transparent, rubric-guided feedback.
 - **Disadvantages:**
 - Limited to theory-based descriptive answers.
 - Performance depends on OCR quality.

- [3] **Agent Design Pattern Catalogue: A Collection of Architectural Patterns for Foundation Model Based Agents**
 - **Authors:** Yue Liu, Sin Kit Lo, Qinghua Lu, Liming Zhu, Dehai Zhao, Xiwei Xu, Stefan Harrer, Jon Whittle
 - **Published in:** The Journal of Systems & Software (Elsevier), 2025
 - **Methodology:**
 - Conducts a **Systematic Literature Review (SLR)** of 57 studies.
 - Proposes **18 architectural design patterns** for Agentic AI systems.
 - Supports **Multi-Agent AI**, **RAG**, planning, reflection, and **Explainable AI (XAI)**.
 - **Accuracy:** No numerical accuracy reported (architecture and design pattern paper).
 - **Advantages:**
 - Provides reusable architectural patterns for Agentic AI systems.
 - Supports Multi-Agent AI, RAG, planning, reflection, and XAI.
 - Helps design scalable and trustworthy AI systems.
 - **Disadvantages:**
 - Focuses on architectural design rather than implementation.
 - Does not evaluate application-specific performance.

- **[4] Machine Learning Models for Academic Performance Prediction: Interpretability and Application in Educational Decision-Making**

- **Authors:** Haiping Wang, Chunhong Zhang, Wenxuan Li, Zhijun Guo and Xiaowei Zhang
- **Published in:** Frontiers in Education, 2025
- **Methodology:**
 - Implemented using **Python**.
 - Utilizes **XGBoost**, **Random Forest**, and **SHAP** for interpretable prediction.
 - Integrates **FastAPI** and **PostgreSQL** for decision support.
- **Accuracy:** XGBoost achieved the best prediction performance with an **F1-score of 0.924**, while SHAP provided transparent feature-level explanations.
- **Advantages:**
 - High prediction accuracy using XGBoost.
 - Explainable predictions through SHAP.
 - Supports data-driven educational decision-making.
- **Disadvantages:**
 - Focused on academic performance prediction rather than document analysis.
 - Requires high-quality educational datasets.

- [5] **Computational Modeling of Educational Excellence: A Systems Approach to Integrating Artificial Intelligence with NAAC Revised Accreditation Framework**
 - **Authors:** Prakash P. and Shivanand S. Gornale
 - **Published in:** 2026
 - **Methodology:**
 - Integrates **AI-based document processing** and **Natural Language Processing (NLP)**.
 - Utilizes **BART, ANN, CNN**, and **RCNN** for NAAC SSR analysis.
 - Generates qualitative score prediction and recommendation-driven reports.
 - **Accuracy:** CNN achieved **98%** accuracy for qualitative score prediction, with criterion extraction accuracy of **94.3%**.
 - **Advantages:**
 - Automates NAAC SSR evaluation and document analysis.
 - Improves prediction accuracy and consistency.
 - Generates recommendation-driven accreditation reports.
 - **Disadvantages:**
 - Focused only on NAAC accreditation documents.
 - Does not support autonomous multi-agent AI.

Literature Survey

Paper	Methodology	Accuracy	Advantages	Disadvantages
A Multi-Agent RAG Framework for Regulatory Compliance Checking of Software Requirements	Multi-Agent RAG framework using LLMs, prompt engineering, XAI and Human-in-the-Loop (HITL) for automated compliance checking	High performance using RAG evaluation metrics	Automated compliance checking, transparent AI decisions and improved accuracy through multi-agent collaboration	High computational complexity; requires multiple AI models and expert validation
Alvaluate: Multi-Agent Framework for Automated Answer Script Evaluation	Multi-agent system using OCR, GPT-4o-mini, rubric-guided evaluation and FAISS	Strong faculty correlation; 70–80% faster evaluation	Fast, consistent grading with transparent feedback	OCR dependent; limited to descriptive answers

Literature Survey (cont.)

Paper	Methodology	Accuracy	Advantages	Disadvantages
Agent Design Pattern Catalogue for Foundation Model Based Agents	Systematic review proposing architectural design patterns supporting planning, reflection, memory, tool usage, RAG and multi-agent collaboration	Not reported	Provides a strong architectural foundation for Agentic AI, RAG, planning and reflection	No implementation or quantitative performance evaluation
Machine Learning Models for Academic Performance Prediction: Interpretability and Application in Educational Decision-Making	Random Forest and XGBoost with SHAP-based interpretability and an interactive decision	$R^2=0.91$; 15% lower MSE	Highly interpretable predictions using SHAP, accurate decision support and interactive	Focused on academic performance prediction; requires large educational datasets

Literature Survey (cont.)

Paper	Methodology	Accuracy	Advantages	Disadvantages
Computational Modeling of Educational Excellence: A Systems Approach to Integrating AI with NAAC	AI document processing using NLP, BART, CNN, ANN and RCNN for automated SSR analysis and report generation	98%	Automates NAAC assessment, document analysis and report generation	Limited to NAAC SSR analysis; does not employ Agentic AI or multi-agent workflows

Functional Requirements

- Allow institutions to upload SSR, AQAR and other accreditation documents.
- Extract text from uploaded PDF and scanned documents using OCR.
- Analyze institutional documents using Agentic AI and Retrieval-Augmented Generation (RAG).
- Detect quality gaps, missing evidence and compliance issues.
- Generate explainable recommendations for accreditation improvement.
- Display quality scores, document analysis and recommendations through an interactive dashboard.

Software Requirements

- Python 3.11.11
- FastAPI
- React.js
- PostgreSQL
- LangChain
- LangGraph
- Gemini 2.5 Flash
- FAISS
- SHAP

- **AI-Powered Document Analysis** – Develop a web-based system to analyze SSR, AQAR and institutional documents using Agentic AI, RAG and LLMs.
- **NAAC Criteria Mapping and Gap Detection** – Map document content to NAAC criteria and identify missing evidence and quality gaps.
- **Explainable Recommendation Generation** – Generate evidence-based recommendations with transparent explanations using Explainable AI (SHAP).
- **Interactive Dashboard and Reporting** – Display quality scores, recommendations and downloadable reports through an interactive dashboard.

- **Document Analysis Agent** – Extracts and analyzes information from uploaded institutional documents.
- **Criteria Mapping Agent** – Maps document content to the relevant NAAC accreditation criteria using RAG.
- **Gap Detection Agent** – Detects missing evidence and quality gaps in institutional documents.
- **Recommendation Agent** – Generates recommendations to improve accreditation readiness.
- **Explanation Agent** – Explains the AI-generated recommendations using SHAP for better transparency.

A feasibility study is conducted to evaluate the practicality and viability of the proposed CampusInsight AI system. The feasibility is analyzed as follows:

Technical Feasibility

- The proposed system will be developed using Python and FastAPI for backend development.
- React.js is proposed to be used for developing an interactive and user-friendly web interface.
- LangGraph, LangChain, Gemini 2.5 Flash, FAISS and SHAP are proposed to support AI-powered document analysis and explainable recommendations.
- The required software tools are open-source or widely available, making the proposed system technically feasible.

Operational Feasibility

- The proposed system will provide a simple web interface for uploading institutional documents.
- Agentic AI is proposed to analyze documents and generate quality insights automatically.
- The dashboard will present quality scores, detected gaps and improvement recommendations.
- SHAP-based explanations will help users understand the reasons behind AI-generated recommendations.
- The proposed system is intended to support faculty members, HODs, principals, and institutional administrators during accreditation activities.

Economic Feasibility

- The proposed system will utilize open-source and freely available technologies, thereby reducing development costs.
- The system is expected to be deployable on a standard computer or institutional server.
- No specialized hardware is anticipated for the proposed implementation.
- The proposed solution is expected to be cost-effective while supporting continuous institutional quality improvement.

- **Step 1: Document Upload**

- Users will upload SSR, AQAR and other institutional quality documents.

- **Step 2: Document Processing**

- The proposed system will preprocess the uploaded documents and prepare them for AI-based analysis.

- **Step 3: AI Document Analysis**

- Agentic AI is proposed to analyze the documents, identify quality gaps and map content to institutional quality criteria using RAG.

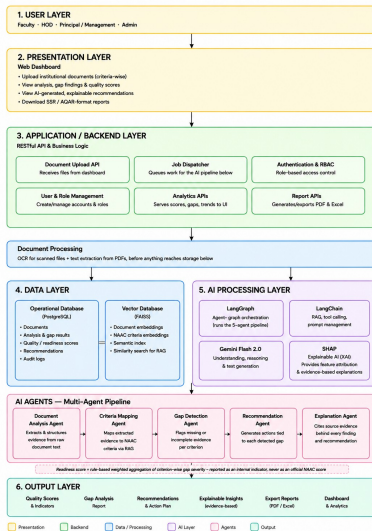
- **Step 4: Recommendation Generation**

- Gemini 2.5 Flash is proposed to generate explainable recommendations and improvement suggestions with SHAP-based explanations.

- **Step 5: Result Visualization**

- The dashboard will display quality scores, detected gaps and AI-generated insights for decision support.

ARCHITECTURE DIAGRAM



USE CASE DIAGRAM

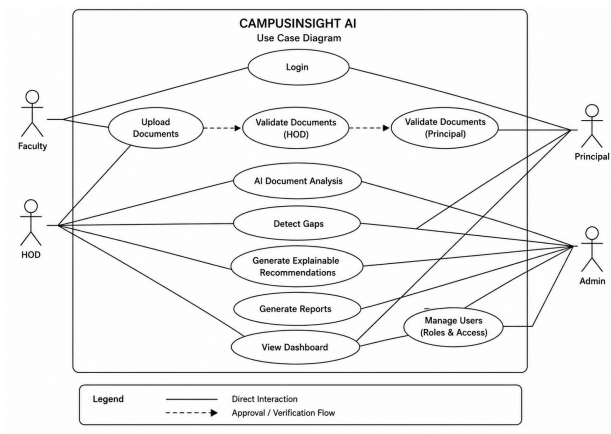


Figure: Use Case Diagram

DATA FLOW DIAGRAM

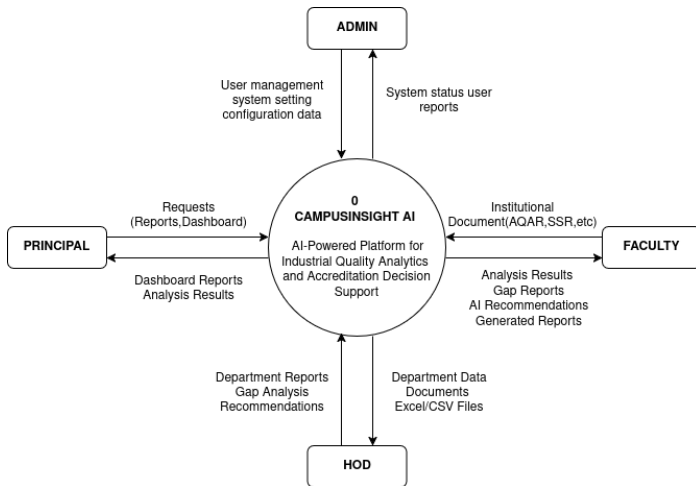


Figure: DFD Level 0

DATA FLOW DIAGRAM

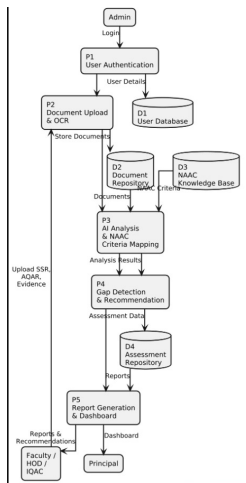
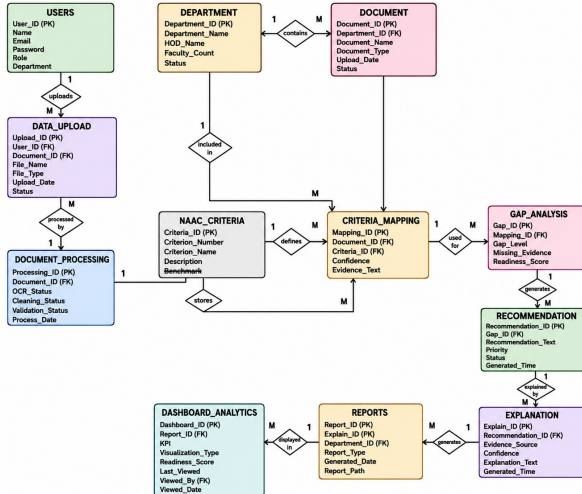


Figure: DFD Level 1

ER DIAGRAM



GANTT CHART

CampusInsight AI — Project Gantt Chart

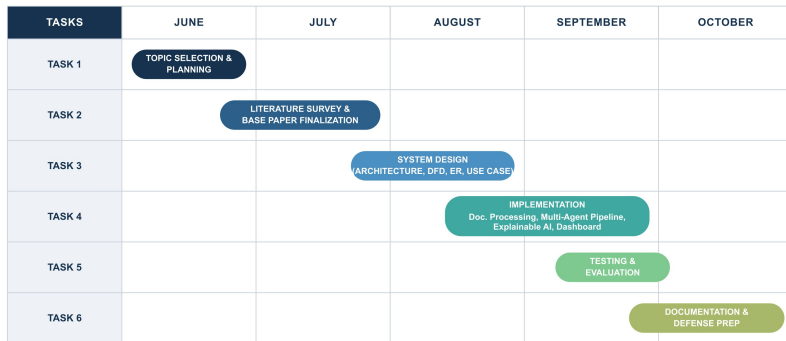


Figure: Gantt Chart

CONCLUSION

- CampusInsight AI is an Agentic AI-based platform that intelligently analyzes institutional quality documents and identifies areas for improvement.
- By integrating Agentic AI, Large Language Models (LLMs), and Explainable AI (XAI), the system provides transparent recommendations and supports informed institutional decision-making.
- The proposed solution helps improve document quality, enhances institutional performance, and supports continuous quality improvement and accreditation readiness.

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<https://doi.org/10.5281/zenodo.16935890>